Rationalising the Denominator with Surds

Exam code: 9MA0

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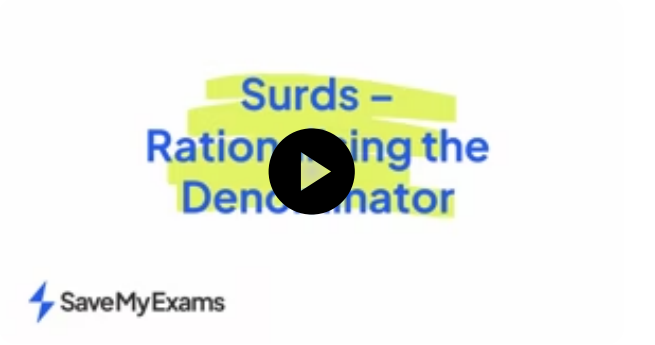

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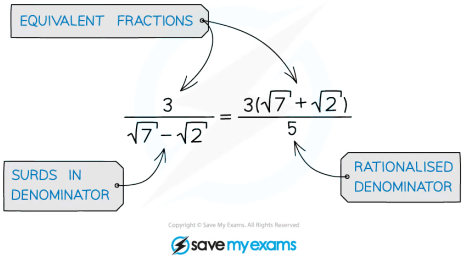
Rationalising the denominator with surds

How do I rationalise the denominator?

- Rationalising a denominator changes a fraction with surds in its denominator, into an equivalent fraction where the denominator is a rational number (usually an integer) and any surds are in the numerator



- There are three cases you need to know how to deal with when rationalising denominators:

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

DENOMINATOR BY $\sqrt{a}$

- IF THE FRACTION IS IN THE FORM $\frac{1}{\sqrt{a}+\sqrt{b}}$, MULTIPLY THE NUMERATOR AND DENOMINATOR BY $\sqrt{a}-\sqrt{b}$
- IF THE FRACTION IS IN THE FORM $\frac{1}{\sqrt{a}-\sqrt{b}}$, MULTIPLY THE NUMERATOR AND DENOMINATOR BY $\sqrt{a}+\sqrt{b}$

IN THE LAST TWO CASES, EITHER $\sqrt{a}$ OR $\sqrt{b}$ CAN REPLACED BY AN INTEGER.
SIMPLY MODIFY YOUR MULTIPLIER TO MATCH.

FOR EXAMPLE, IF YOUR FRACTION IS IN THE FORM $\frac{1}{a+\sqrt{b}}$ THEN MULTIPLY THE NUMERATOR AND DENOMINATOR BY $a-\sqrt{b}$.

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Examiner Tips and Tricks

- If an exam question asks you to give an answer, for example, "in the form $p + q\sqrt{3}$, where p and q are rational numbers", this does NOT mean that p and q have to be integers, or positive!
- Remember: both integers and fractions (both positive and negative) are rational numbers

Worked Example

where p and q are rational numbers.

USE FOIL TO MULTIPLY THESE TERMS OUT

$3\sqrt{3} \times \sqrt{3}$
$= 3\sqrt{9}$
$= 9$

$$\frac{4-3\sqrt{3}}{5-\sqrt{3}} \times \frac{5+\sqrt{3}}{5+\sqrt{3}} = \frac{20+4\sqrt{3}-15\sqrt{3}-9}{25-3}$$

DIFFERENCE OF SQUARES, WITH $(\sqrt{3})^2 = 3$

$$= \frac{11-11\sqrt{3}}{22}$$

$$= \frac{11}{22} - \frac{11\sqrt{3}}{22}$$

$$= \frac{1}{2} - \frac{1}{2}\sqrt{3}$$

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